High Torque Timing Pulleys - S3M

Compatible with S2M type from Mitsuboshi Belting Ltd. as well as Bando Chemical Industries Ltd.

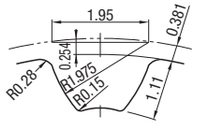
■For High Torque Timing Belts, see P.1465 and for Idlers with Teeth, see P.1431. For High Torque Timing Belts, see P.1449.



Type	Belt Width 6mm S3M060	Belt Width 10mm S3M100	Belt Width 15mm S3M150	M Material *1 Pulley	M Material *1 Flange	S Surface Treatment	A Accessory *1 Set Screws
HTPA	●	●	●	A2000 Series Aluminum Alloy	Aluminum Alloy	Clear Anodize	EN 1.4301 Equiv.
HTPB	●	●	●			Black Anodize	
HTPK	●	●	●			Hard Clear Anodize *2	
HTPN	●	●	●			Electroless Nickel Plating	
HTPT	●	●	●	EN 1.1191 Equiv.	EN 1.0330 Equiv.	-	EN 1.7220 Equiv. (Black Oxide)
HTPM	●	●	●			Black Oxide	
HTPP	●	●	●			Electroless Nickel Plating	
HTPS	●	●	-	EN 1.4301 Equiv.	EN 1.4301 Equiv.	-	EN 1.4301 Equiv.

Flange is installed, and set screws are included with Shaft Bores P, N and C. *1. The above material and accessory might be changed to the ones equivalent to the originals. *2. Hard Clear Anodize: Film Hardness 300HV ~

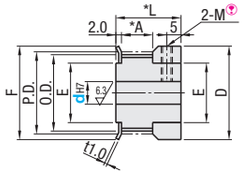
Standard Tooth Profile



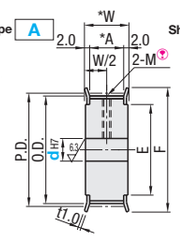
Tooth groove dimensions slightly vary according to the number of teeth.
(Pitch: 3.0mm)

• Pulley Shape

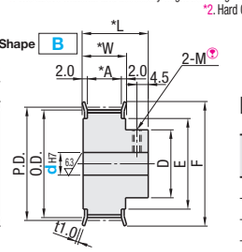
Shape K



Shape A



Shape B



Shaft Bore Specs. H (Round hole), V or F (Stepped Hole) and Y (Both Sides Stepped Hole) do not have tapped holes.

■Tapped Hole Dimensions (Shaft Bore Specs.: P, N, C)

d H7 Shaft Bore I.D.	M (Coarse)	Accessory: Set Screw
4~5	M3	M3x3
6~17	M4	M4x3
18~33	M5	M5x4
34~42	M6	M6x5

■Number of Teeth / Dimension

mm	Number of Teeth 14	15	16	18	19	20	22	24	25	26	28	30	32	34	36	40	44	48	50	60	72
P.D.	13.37	14.32	15.28	17.19	18.14	19.10	21.01	22.92	23.87	24.83	26.74	28.65	30.56	32.47	34.38	38.20	42.02	45.84	47.75	57.30	68.75
O.D.	12.61	13.56	14.52	16.43	17.38	18.34	20.25	22.16	23.11	24.07	25.98	27.89	29.80	31.71	33.62	37.44	41.25	45.07	46.98	56.53	67.99
D	16	18	18	20	22	22	25	14	16	16	18	20	20	26	26	30	32	34	34	39	50
F	16	18	18	20	22	22	25	25	28	28	30	32	35	40	40	44	48	50	52	61	74
E	10	11	11	13	14	14	16	16	18	18	20	23	25	28	28	32	36	38	40	46	58

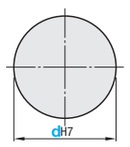
■Belt Nominal Width / Dimension

mm	Nominal S3M060	S3M100	S3M150
A	7	11	17
W	11	15	21
L	19	23	29

Shaft Bore Specs

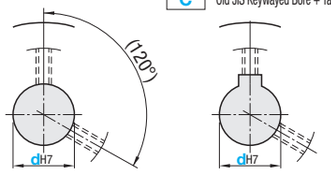
Surface treatment may not be applied to shaft bores.

H Round Hole



No tapped holes or set screws.

P Round Hole + Tap

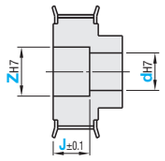


For A-Shape pulley, the screw holes are set at around 120° to keep away from peaks.
Shaft Bore Specs. P is not available for Shape A with 22 or less teeth and nominal width 060.

N New JIS Keywayed Bore + Tap
C Old JIS Keywayed Bore + Tap

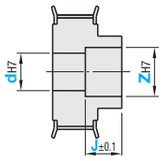
For Keyway Dimension Details, see P.1377.
When selecting the shaft bore dia. 10 and the keyway width 4.0mm (height 1.8mm) for New JIS Keywayed Bore, specify NK10.

V Stepped Hole



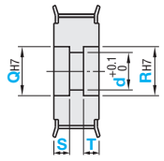
Not applicable to Shape K.
No tapped holes or set screws.

F Stepped Hole (Counterbored Holes on the Hub Side)



Applicable to Shape B only.
No tapped holes or set screws.

Y Both Ends Stepped Hole



Applicable to Shape A only.
Shaft Bore Dia. d is +0.1 / 0
No tapped holes or set screws.

Part Number: Type	Number of Teeth	Type Nominal Width	Pulley Shape	Pulley Shape: A — Shaft Bore Specifications (~): Specify in 1mm Increment, (,): Select the former or latter — H Round Hole	P Round Hole + Tap * See the paragraph marked with ⊗	N, C Keyway + Tap	V Stepped Hole V	V Stepped Hole Z-d≥2	V Stepped Hole J (0.1mm Increment)	Y Both Ends Stepped Y	Y Both Ends Stepped Q, R Q(R)-d≥2	Y Both Ends Stepped S, T
Aluminum HTPA HTPB HTPK HTPN	14		A	4, 5, 6	-	-	4	6	2.0≤J≤W-2.0	4	6	3~9 S+T≤W-3
	15			4, 5, 6	-		4	6		4	6	
	16			4~6, 6.35, 7	-		4, 5	6, 7		4, 5	6, 7	
	18			4~6, 6.35, 7, 8	4, 5		4, 5, 6	6, 7, 8		4, 5, 6	6, 7, 8	
	19			4~6, 6.35, 7~9	4, 5, 6		4~7	6~9		4~7	6~9	
	20			4~6, 6.35, 7~9	4, 5, 6		4~7	6~9		4~7	6~9	
	22			4~6, 6.35, 7~10	4~6, 6.35, 7, 8		4~7	6~9		4~7	6~9	
Steel HTPT HTPM HTPP	24	S3M060	K	4~6, 6.35, 7~10	4~6, 6.35, 7~10	-	4~7	6~9		4~7	6~9	
	25			4~6, 6.35, 7~11	4~6, 6.35, 7~11	8, 10, NK10, 11	4~8	6~11		4~8	6~11	
	26			5, 6, 6.35, 7~11	5, 6, 6.35, 7~11	8, 10, NK10, 11	5~8	7~11		5~8	7~11	
	28	S3M100		5, 6, 6.35, 7~13	5, 6, 6.35, 7~13	8, 10, NK10, 11~13	5~10	7~12		5~10	7~12	
	30			6, 6.35, 7~15	6, 6.35, 7~14	8, 10, NK10, 11~14	6~13	8~15		6~13	8~15	
	32	S3M150		6, 6.35, 7~17	6, 6.35, 7~16	8, 10, NK10, 11~16	6~15	8~17		6~15	8~17	
	34		A	6, 6.35, 7~20	6, 6.35, 7~18	8, 10, NK10, 11~18	6~18	8~23		6~18	8~20	
Stainless Steel HTPS	36		B	6, 6.35, 7~20	6, 6.35, 7~18	8, 10, NK10, 11~18	6~18	8~23		6~18	8~20	
	40			8~24	8~23	8, 10, NK10, 11~23	8~22	10~24		8~22	10~24	
	44			8~28	8~25	8, 10, NK10, 11~25	8~26	10~28		8~26	10~28	
	48			8~30	8~25	8, 10, NK10, 11~25	8~28	10~30		8~28	10~30	
	50			8~32	8~28	8, 10, NK10, 11~28	8~30	10~32		8~30	10~32	
	60			8~38	8~33	8, 10, NK10, 11~32	8~36	10~38		8~36	10~38	
	72			8~50	8~42	8, 10, NK10, 11~40	8~48	10~50		8~48	10~50	

Part Number: Type	Number of Teeth	Type Nominal Width	Pulley Shape	Pulley Shape: B, K — Shaft Bore Specifications (~): Specify in 1mm Increment, (,): Select the former or latter — H Round Hole	P Round Hole + Tap	N, C Keyway + Tap	V, F Stepped Hole V, F	V, F Stepped Hole Z-d≥2	V, F Stepped Hole J (0.1mm Increment)
Aluminum HTPA HTPB HTPK HTPN	14		A	4, 5, 6	4, 5, 6				
	15			4, 5, 6	4, 5, 6				
	16			4~6, 6.35, 7	4~6, 6.35, 7				
	18			4~6, 6.35, 7, 8	4~6, 6.35, 7, 8	-	-	-	
	19			4~6, 6.35, 7~9	4~6, 6.35, 7, 8				
	20			4~6, 6.35, 7~9	4~6, 6.35, 7, 8				
	22			4~6, 6.35, 7~10	4~6, 6.35, 7~10				
Steel HTPT HTPM HTPP	24	S3M060	K	4~6, 6.35, 7~10	4~6, 6.35, 7, 8	-	4~6	6~8	
	25			4~6, 6.35, 7~11	4~6, 6.35, 7~10	8	4~8	6~10	
	26			5, 6, 6.35, 7~11	5, 6, 6.35, 7~10	8	5~8	7~10	
	28	S3M100		5, 6, 6.35, 7~13	5, 6, 6.35, 7~10	8	5~10	7~12	2.0≤J≤L-2.0
	30			6, 6.35, 7~15	6, 6.35, 7~12	8, 10, NK10, 11	6~13	8~15	
	32	S3M150		6, 6.35, 7~16	6, 6.35, 7~12	8, 10, NK10, 11	6~14	8~16	
	34		A	6, 6.35, 7~20	6, 6.35, 7~18	8, 10, NK10, 11~13	6~18	8~23	
Stainless Steel HTPS	36		B	6, 6.35, 7~20	6, 6.35, 7~18	8, 10, NK10, 11~13	6~18	8~23	
	40			8~24	8~21	8, 10, NK10, 11~16	8~22	10~24	
	44			8~28	8~23	8, 10, NK10, 11~18	8~26	10~28	
	48			8~30	8~25	8, 10, NK10, 11~20	8~28	10~30	
	50			8~30	8~25	8, 10, NK10, 11~20	8~28	10~30	
	60			8~35	8~30	8, 10, NK10, 11~22	8~33	10~35	
	72			8~46	8~38	8, 10, NK10, 11~32	8~44	10~46	

⊗Shaft Bore Dia. 8, 11, 13, 14, 17, 21~32 are not available for Shaft Bore Spec. C. ⊗Shaft Bore Specs. P is not available for Shape A with 22 or less teeth and nominal width S3M060.

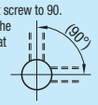
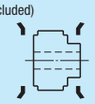
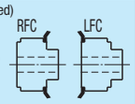
Ordering Example

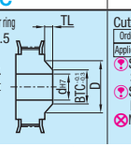
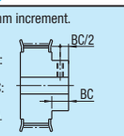
	Part Number	Pulley Shape	Shaft Bore Specs., I.D.	Z	J	Q	R	S	T
(Shaft Bore Specs.: H, P, N, C)	HTPA32S3M100 -	A -	C16						
(Shaft Bore Specs.: V, F)	HTPA48S3M150 -	A -	V12 -	Z26 -	J17.0				
(Shaft Bore Specs: Y)	HTPA36S3M100 -	A -	Y10			- Q19 -	R15 -	S5 -	T4

Number of Teeth	Body Price HTPA (x1.0) HTPB, HTPK (x1.1) HTPN (x1.2) S3M060 Shape A	S3M060 Shapes B and K	S3M100 Shape A	S3M100 Shapes B and K	S3M150 Shape A	S3M150 Shapes B and K	HTPT (x1.0) HTPM (x1.05) HTPP (x1.15) S3M060 Shape A	S3M060 Shapes B and K	S3M100 Shape A	S3M100 Shapes B and K	S3M150 Shape A	S3M150 Shapes B and K	HTPS (x1.0) S3M060 Shape A	S3M060 Shapes B and K	S3M100 Shape A	S3M100 Shapes B and K	Shaft Bore Machining Charge (Body Price +) HTPA, HTPB HTPK, HTPN P Hole	N, C, V, F Hole	Y Hole	HTPT, HTPM, HTPP P Hole	N, C, V, F Hole	Y Hole	HTPS P Hole	N, C, V, F Hole	Y Hole
14																									
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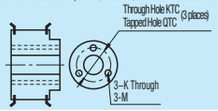
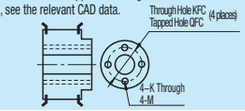
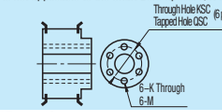
Alterations

Part Number	Pulley Shape	Shaft Bore Specs., I.D.	Z	J	Q	R	S	T	(KC90···etc.)
HTPA60S3M100 -	A -	H16						-	KSC25 - K4

Alterations	Set Screw Angle	Flange Not Swaged	Flange Swaged on One Side	Flange Cut
Code	KC90	NFC	RFC, LFC	FC
Spec.	Changes an angle of set screw to 90. For A-Shape pulley, the screw holes are set at around 90° to keep away from peaks. 	(Flange 2 pcs. Included) Ordering Code NFC 	(Flange 1 pc. Included) Ordering Code RFC Application Notes ⊗Not applicable to Shape K. 	Cut the flange O.D. in 0.5mm increment. Ordering Code FC17 Application Notes FC≥(O. D.)+1 FC≤F-2 No surface treatment is applied on flange circumference. ⊗Not available for Stainless Steel Type. 

Alterations	Adds taper for retaining bearing	Hub Shortening	Tapped Hole Dimensions	Changes the length of the included set screws.
Code	BTC	BC	TPC	SLH
Spec.	Add taper for retaining bearing inner ring Ordering Code BTC4-TL1.5 Application Notes Applicable to Shape A only. Applicable to Shaft Bore Specs. H and P only. TL<L-W 	Cuts the hub length in 0.5mm increment. Ordering Code BC6.5 Application Notes Shaft Bore Specs. H, V, F: 3≤BC≤L-W Shaft Bore Specs. P, N, C: M+3≤BC≤L-W ⊗Not available for Shapes K and A. 	Ordering Code TPC5 Application Notes Applicable to Shaft Bore Specs. P, N, C only. M: TPC M3: M4 M4: M3, M5 M5: M4	Ordering Code SLH10 Application Notes Applicable to Shaft Bore Specs. P, N, C only. Set Screws: SLH M3x3: 6 M4x3: 5, 8 M5x4: 6, 10 M6x5: 10

For details, see the "Timing Pulley Alterations - Overview" section (P.1378).

Alterations	Side Through Hole / Side Tapped Hole, 3 places	Side Through Hole / Side Tapped Hole, 4 places	Side Through Hole / Side Tapped Hole, 6 places
Code	KTC, QTC	KFC, QFC	KSC, QSC
Spec.	Machine Through Hole / Tapped Hole on the side surface of hub side Ordering Code (Through Hole) KTC20-K5.0 Ordering Code (Tapped Hole) QTC28-M4 Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment) Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8 Application Notes ⊗Not applicable to Shape K. ⊗Not applicable to Shaft Bore Specs. F or Y. ⊗When KTC/QTC is selected for Shaft Bore Specs. P, N and C, KC90 is not available. 	Machine Through Hole / Tapped Hole on the side surface of hub side Ordering Code (Through Hole) KFC20-K5.0 Ordering Code (Tapped Hole) QFC28-M4 Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment) Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8 Application Notes ⊗Not applicable to Shape K. ⊗Not applicable to Shaft Bore Specs. F or Y. Specify KC90 when selecting KFC/QFC for Shaft Bore Specs. P, N and C. Side holes and tooth side tapped holes might interfere with each other. For details, see the relevant CAD data. 	Machine Through Hole / Tapped Hole on the side surface of hub side Ordering Code (Through Hole) KSC20-K5.0 Ordering Code (Tapped Hole) QSC28-M4 Selection (Through Hole) K Selection K4.0~K13.0 (0.5mm Increment) Selection (Tapped Hole) M Selection M3, M4, M5, M6, M8 Application Notes ⊗Not applicable to Shape K. ⊗Not applicable to Shaft Bore Specs. F or Y. ⊗KSC/QSC is not applicable to the Shaft Bore Specs. P, N and C. 

For details, see the "Timing Pulley Alterations - Overview" section (P.1378).